Thoracic Respiratory Tract & Lungs

1. The trachea enters the superior mediastinum and descends to the sternal angle, where it divides into the right and left primary bronchi. This critical anatomical bifurcation occurs at which vertebral level in a recumbent subject?
a. Lower border of the 4th thoracic vertebra (T4/T5 disc level)
b. Spinous process of the T2 vertebra
c. Upper border of the T7 vertebra
d. Xiphisternal joint plane
e. Inferior thoracic aperture boundary
2. The right and left primary (main) bronchi differ significantly in their structural configuration. Compared to the left main bronchus, the right main bronchus is characteristically:
a. Shorter, wider, and oriented more vertically (in direct alignment with the trachea)
b. Longer, narrower, and oriented horizontally
c. Completely devoid of any structural cartilage support
d. Wrapped tightly inside the arch of the aorta loop
e. Positioned completely posterior to the esophagus
3. Because of the vertical alignment and greater diameter of the right primary bronchus, inhaled foreign bodies or dental debris aspirated during clinical procedures are most likely to lodge in:
a. The left upper lung lobe
b. The right lung or its lower lobar structures
c. The terminal submeniscal recesses
d. The walls of the left terminal bronchiole
e. The horizontal segment of the thoracic duct
4. The lungs are conical organs suspended within the thoracic cavity. Each lung is covered by a delicate, double-layered serous membrane shell known as the:
a. Pericardium
b. Peritoneum
c. Endothoracic fascia
d. Pleura (visceral and parietal layers)
e. Suprapleural membrane
5. The visceral pleura is completely insensitive to ordinary somatic pain because it lacks somatic innervation; instead, it is supplied by autonomic fibers. It is structurally continuous with the parietal pleura at the:
a. Jugular notch
b. Sternal angle
c. Scleral spur area
d. Root of the lung (hilum)
e. Costodiaphragmatic recess base
6. The potential space between the visceral and parietal pleura contains a microscopic film of serous fluid that reduces friction. Accumulation of excess fluid in this space due to inflammation or infection is a:
a. Pneumothorax
b. Pleural effusion
c. Cardiac tamponade

d. Goiter compression
e. Bronchial mucocele

7. Air entering the pleural cavity due to a penetrating chest wound destroys the natural negative intrapleural pressure, causing the underlying lung to collapse. This life-threatening condition is a:

a. Chylothorax
b. Hemothorax
c. Pneumothorax
d. Hydrocephalus shift
e. Trismus crisis

8. The right lung is larger and heavier than the left lung. Macroscopically, it is divided into superior, middle, and inferior lobes by which two anatomical fissures?

a. Horizontal and oblique fissures
b. Vertical and transverse clefts
c. Sagittal and coronal sulci
d. Marginal and mental borders
e. Superior and inferior sinus slots

9. The left lung is smaller than the right lung because of the asymmetrical space occupied by the heart. It is divided into superior and inferior lobes by a single oblique fissure, and its anterior border features a distinct indentation named the:

a. Cardiac notch (with the adjacent tongue-like lingula)
b. Hilum recess
c. Sulcus terminalis
d. Costal margin
e. Bronchopulmonary groove

10. The medial surface of each lung features a wedge-shaped area where bronchi, pulmonary vessels, lymphatics, and nerves enter and leave the organ. This high-yield structural gate is the:

a. Carina
b. Apex
c. Base
d. Hilum
e. Lingula

11. Within the hilum of the right lung, the primary bronchus divides early. The branch that passes *above* the pulmonary artery is designated as the:

a. Hyparterial bronchus
b. Eparterial bronchus
c. Terminal bronchiole
d. Respiratory duct
e. Lobar artery

12. Within both lung hila, the primary bronchus branches that pass *below* the level of the main pulmonary artery are termed the:

a. Eparterial bronchi
b. Hyparterial bronchi
c. Segmental carinae
d. Alveolar sacs
e. Trachealis muscle slips

13. The lungs are divided into structurally and functionally independent anatomical units, each supplied by a tertiary (segmental) bronchus and a dedicated branch of the pulmonary artery. These surgically resectable units are the:
a. Pulmonary lobules
b. Acini
c. Bronchopulmonary segments (10 in the right lung, 8–10 in the left lung)
d. Alveolar units
e. Pleural recesses
14. Deoxygenated blood is carried from the right ventricle of the heart to the lungs for gas exchange by the pulmonary arteries. Oxygenated blood is then returned to the left atrium of the heart via the:
a. Superior vena cava
b. Bronchial veins
c. Pulmonary veins (two from each lung)
d. Azygos vein system
e. Internal thoracic veins
15. The structural lung tissue (parenchyma, bronchi, and visceral pleura) receives its high-pressure oxygenated nutritional blood supply directly from the systemic circulation via the:
a. Pulmonary arteries
b. Bronchial arteries (branches of the descending thoracic aorta)
c. Internal thoracic arteries
d. Coronary arteries
e. Superior thyroid arteries
16. Histologically, the conducting portion of the respiratory tract (including the primary, secondary, and tertiary bronchi) is lined by a specialized mucosal epithelium. This respiratory epithelium is characterized as:
a. Pseudostratified ciliated columnar epithelium with goblet cells
b. Stratified squamous non-keratinized epithelium
c. Simple squamous epithelium
d. Transitional cell epithelium
e. Simple cuboidal brush border matrix
17. As the bronchial tree branches deeper into the lung parenchyma and decreases in diameter, the tall pseudostratified ciliated columnar epithelial lining gradually transitions into:
a. Keratinized stratified squamous scales
b. Simple columnar and then simple cuboidal epithelium
c. Stratified cuboidal cells
d. Stratified columnar blocks
e. Transitional rows
18. The walls of the primary and secondary bronchi feature structural support elements that keep the airway patent. Under the microscope, these support elements consist of:
a. Continuous, rigid closed bone rings
b. Irregular plates and islands of hyaline cartilage
c. Pure skeletal muscle sheets lacking cartilage
d. Intercalated collagen cords
e. Concentric zymogen deposits

19. Microscopic evaluation of a bronchus wall reveals a distinct layer of smooth muscle situated between the lamina propria and the cartilage plates. This smooth muscle is arranged:
a. In two opposing vertical vertical columns
b. Spirally or circularly around the lumen
c. In random unmapped chaotic patches
d. Only along the posterior boundary line
e. Interspersed within the cartilage matrix
20. When the bronchial tree branches down to a diameter of approximately 1 mm or less, the cartilage plates and mucosal glands disappear completely. These micro-anatomical segments are the:
a. Primary bronchi
b. Secondary lobar bronchi
c. Bronchioles
d. Alveolar ducts
e. Atria
21. Because bronchioles completely lack cartilage plates to maintain their patency, their lumen diameter is regulated entirely by the contraction or relaxation of their thick wall component of:
a. Hyaline fibers
b. Elastic ligaments
c. Smooth muscle (under autonomic nervous system control)
d. Stratified squamous layers
e. Calcified sand tracks
22. Microscopic analysis of the simple cuboidal epithelium lining the bronchioles reveals a population of non-ciliated, dome-shaped cells with apical secretory granules. These high-yield cells are the:
a. Goblet cells
b. Kulchitsky cells
c. Club cells (formerly Clara cells)
d. Type I pneumocytes
e. Myoepithelial units
23. Club cells (Clara cells) provide essential protective functions. Histologically, they synthesize and secrete glycosaminoglycans, cytochrome P450 enzymes to detoxify inhaled toxins, and:
a. Pure hydrochloric acid
b. Surfactant-like lipoproteins to keep the small airways open
c. Zymogen digestive polymers
d. Melanin granules
e. Mucin complexes
24. The terminal bronchioles are the final segments of the purely conducting portion of the respiratory tract. They branch directly into the:
a. Secondary bronchi
b. Segmental bronchi
c. Alveolar sacs
d. Respiratory bronchioles (marking the start of the respiratory division)
e. Atria

25. Respiratory bronchioles are structurally unique under the microscope because their walls are interrupted by the presence of occasional, thin-walled saccular outpocketings called:
a. Bronchial glands
b. Mucous crypts
c. Alveoli
d. Lymphoid follicles
e. Vascular demilunes
26. Respiratory bronchioles branch into long, linear corridors lined completely by a series of continuous alveoli. These microscopic corridors are the:
a. Terminal channels
b. Alveolar ducts
c. Alveolar sacs
d. Interlobular septa
e. Fibrous trabeculae
27. Alveolar ducts terminate as dead-end clusters where multiple individual alveoli open into a shared central space. These specialized clusters are the:
a. Atria
b. Alveolar sacs
c. Respiratory acini
d. Terminal rings
e. Bronchial bulbs
28. The specialized wall or partition situated between two adjacent individual alveoli, containing an exceptionally rich network of continuous capillaries supported by elastic and reticular fibers, is the:
a. Interlobular septum
b. Visceral pleura sheet
c. Interalveolar septum
d. Adventitia capsule
e. Basement lamina proper
29. The internal surface of each alveolus is lined by a highly delicate alveolar epithelium composed of two main cell types. The extremely thin, flat, squamous cells that cover roughly 95% of the alveolar surface area are the:
a. Type II pneumocytes
b. Alveolar macrophages (dust cells)
c. Type I pneumocytes (squamous alveolar cells)
d. Endothelial cells
e. Goblet units
30. Microscopic evaluation confirms that Type I pneumocytes are bound tightly to each other and to adjacent cells by occluding junctions. Their primary physiological role is to:
a. Synthesize surfactant polymers
b. Act as a barrier and facilitate the rapid diffusion of gases
c. Phagocytose inhaled carbon dust particles
d. Undergo continuous active mitotic replication
e. Secrete mucus films

31. Type II pneumocytes (great alveolar cells) are cuboidal cells scattered among the Type I cells, often located at the junctions of alveolar septa. Under the microscope, their cytoplasm is packed with unique vesicular structures called:
a. Zymogen granules
b. Weibel-Palade bodies
c. Multilamellar bodies (lamellar bodies)
d. Nissl bodies
e. Keratohyalin pools
32. The lamellar bodies of Type II pneumocytes contain concentric sheets of phospholipids and proteins that are secreted onto the alveolar surface to form:
a. A dense mucus plug
b. Pulmonary surfactant (which lowers surface tension to prevent alveolar collapse)
c. A thick calcified shell
d. A high-pressure capillary barrier
e. An enzymatic barrier to trap bacteria
33. In premature infants born before the Type II pneumocytes are fully mature and functional (typically before the 28th to 32nd week of gestation), a severe deficiency of pulmonary surfactant leads to:
a. Asthma
b. Emphysema
c. Infant Respiratory Distress Syndrome (IRDS / Hyaline Membrane Disease)
d. Bronchiectasis
e. Congenital goiter collapse
34. Migrating freely over the internal epithelial surface within the alveolar lumen are large, phagocytic cells derived from blood monocytes. These are the:
a. Type I pneumocytes
b. Type II pneumocytes
c. Alveolar macrophages (dust cells)
d. Langerhans cells
e. Plasma cells
35. In patients with chronic congestive heart failure, high venous pressure causes red blood cells to escape into the alveolar spaces, where they are phagocytosed by alveolar macrophages. These brown, hemosiderin-laden cells are clinically called:
a. Heart failure cells
b. C-cells
c. Kulchitsky cells
d. Basket cells
e. Pituicytes
36. The blood-air barrier (respiratory membrane) represents the minimal physical thickness through which oxygen and carbon dioxide must diffuse between the air and blood. From the alveolar lumen to the capillary lumen, this barrier consists of:
a. Visceral pleura, endothoracic fascia, and parietal pleura
b. Surfactant, Type I pneumocyte, fused basal laminae, and capillary endothelial cell
c. Smooth muscle, elastic cartilage, and adventitia
d. Mucus film, pseudostratified epithelium, and lamina propria
e. Club cell cytoplasm and smooth muscle basement membrane

37. Pathological destruction of the interalveolar septa and loss of elastic fibers, leading to abnormal, permanent enlargement of the air spaces distal to the terminal bronchioles and air trapping, is:
a. Chronic bronchitis
b. Bronchial asthma
c. Emphysema
d. Atelectasis
e. Pneumonia
38. Hyper-reactivity of the tracheobronchial tree smooth muscle, leading to paroxysmal bronchospasm, mucosal edema, and hypersecretion of thick mucus following exposure to allergens, is the clinical hallmark of:
a. Emphysema
b. Bronchial asthma
c. Pleural effusion
d. Pulmonary embolism
e. Respiratory distress syndrome
39. Microscopic evaluation of a lung biopsy from a patient with severe, unmanaged pneumonia reveals that the alveolar lumens are completely filled with a dense inflammatory exudate composed of:
a. Clear cerebrospinal fluid
b. Pure surfactant fluid loops
c. Neutrophils, fibrin, and cellular debris
d. Trapped chondrocytes and osteoclasts
e. Stratified squamous non-keratinized nests
40. The structural nerve supply to the lungs and bronchial tree is mediated via the pulmonary plexuses located at the roots of the organs. Parasympathetic preganglionic fibers are carried to these plexuses by the:
a. Phrenic nerves
b. Sympathetic trunk ganglia
c. Vagus nerves (CN X)
d. Intercostal nerves
e. Recurrent laryngeal nerves
41. Activation of the parasympathetic fibers from the vagus nerve inside the lung parenchyma causes a coordinated physiological response characterized by:
a. Bronchodilation and decreased secretion
b. Bronchoconstriction and increased mucus secretion
c. Complete atrophy of the cartilage plates
d. Shifting of the lung roots superiorly
e. Shutdown of alveolar macrophage phagocytosis
42. Sympathetic postganglionic fibers enter the pulmonary plexuses from the upper thoracic sympathetic chain. Stimulation of these sympathetic pathways causes:
a. Severe bronchospasm
b. Bronchodilation (relaxation of bronchial smooth muscle) and vasoconstriction
c. Profuse salivation from the respiratory tract mucosa
d. Acceleration of surfactant breakdown
e. Immediate collapse of the visceral pleura

43. Sensory (afferent) fibers from the stretch receptors inside the alveolar walls and bronchial smooth muscle travel back to the brainstem via the vagus nerve. These fibers drive which regulatory respiratory reflex?
a. Gag reflex
b. Hering-Breuer reflex (which limits over-inflation of the lungs)
c. Cough reflex via the internal laryngeal nerve
d. Corneal blink reflex
e. Pupillary light reflex consensual response
44. The superficial lymphatic plexus of the lung is located immediately deep to which macroscopic structure, draining its fluid toward the bronchopulmonary lymph nodes?
a. The fibrous pericardium
b. The costal cartilages table
c. The visceral pleura
d. The trachealis muscle core
e. The xiphoid process lamina
45. Deep lymphatic drainage tracks along the paths of the bronchi and pulmonary vessels, passing through the pulmonary and bronchopulmonary (hilar) nodes to empty into the:
a. Submental lymph nodes
b. Tracheobronchial and brachiocephalic lymph nodes, leading to the thoracic or right lymphatic ducts
c. Retropharyngeal lymph nodes
d. Mastoid node groups
e. Superficial cervical chain channels

Answer Key Table (Thoracic Respiratory Tract & Lungs)

Q 1-8	Q 9-16	Q 17-24	Q 25-32	Q 33-40	Q 41-45
a	a	b	c	c	b
a	d	b	b	a	b
b	b	b	b	b	b
d	c	c	b	c	c
d	b	c	b	c	b
b	c	b	b	b	
c	c	a	b	c	
a	b	d	b	c	